

KRISHNA TUSHAR VAGHELA

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PROFESSIONAL SUMMARY

ML/AI Engineer with demonstrated expertise in computer vision and full-stack development. Built and deployed three production-grade systems with documented performance metrics: PAN Card OCR (94% accuracy), Voter ID field extraction (97%+ confidence), and QR-based authentication. Proven ability to implement end-to-end pipelines from data collection through model optimization to deployment. Proficient in PyTorch, YOLO, FPGA design, and industrial IoT systems.

PROFESSIONAL EXPERIENCE

AI/ML Intern – Computer Vision & OCR | Suntek Energy Systems Pvt. Ltd.

Jun 2025 – Nov 2025 • Mumbai, India

- Built YOLOv8-based OCR detection pipelines for document processing, achieving 94.2% MAP on curated datasets.
- Implemented automated field extraction for multi-field documents using EasyOCR with regex validation, cutting manual data-entry effort by 75%.
- Developed end-to-end inference pipelines with confidence scoring, processing 1,000+ documents/month at 97%+ confidence.
- Applied transfer learning, data augmentation, and quantization for production deployment.

Research & Development Intern | Jawaharlal Nehru Port Authority (JNPA)

Jan 2023 – Jun 2023 • Uran, Mumbai

- Contributed to port operations research covering IoT infrastructure and SCADA platforms.
- Produced technical documentation and analysis for operational and data management systems.

DEPLOYED PROJECTS & RESEARCH

RAGmind - Intelligent Document Intelligence Platform [\[GitHub\]](#) [\[Research Paper\]](#)

RAGmind, an open-source, locally-deployed Retrieval-Augmented Generation (RAG) platform that enables intelligent, conversational access to private PDF documents without transmitting data to external servers. RAGmind integrates a Flask-based REST API, HuggingFace sentence-transformer embeddings, ChromaDB vector storage, and a locally-hosted Mistral-7B large language model via Ollama. The system supports multi-turn conversation with persistent session context, returning grounded answers with page-level citations.

PAN Card OCR Detection & Extraction System [\[GitHub\]](#) [\[Research Paper\]](#)

Built end-to-end OCR pipeline achieving 94.2% mean Average Precision (MAP) on custom PAN card dataset. Trained YOLOv8 detector on 71-image curated dataset with augmentation; integrated EasyOCR for character recognition (92% accuracy). Implemented post-processing with PAN regex validation outputs structured JSON/CSV for downstream workflows. Documented technical approach and results in peer-reviewed research format with performance curves and metrics.

Automated Voter ID Field Extraction System [\[GitHub\]](#) [\[Research Paper\]](#)

Designed multi-class detector using YOLOv8 to localize 14 key voter ID fields across varied image quality and orientations. Trained on custom annotated dataset of ~1,284 images; achieved 97%+ average confidence scores for field localization. Integrated EasyOCR for text extraction with confidence scoring and regex validation; built inference pipeline automating document digitization

and reducing manual data-entry effort by ~75%. Published results with detailed performance visualizations

QR Code Authentication System [\[Live Demo\]](#)

Full-stack web application using PHP backend and MySQL database enabling QR-based user authentication. Implemented user registration with dynamic QR code generation, scanning/detection pipeline, and real-time validation. Database-backed verification system with user authentication flow. Deployed with UI demonstrating practical use cases including registration forms, QR generation, detection, and successful login feedback.

SHA-3 (Keccak) FPGA Implementation [Verilog/VHDL]

Implemented hardware SHA-3 cryptographic engine on FPGA with pipelined Keccak-f permutation architecture. Performed RTL design and functional validation using Verilog/VHDL. Profiled and optimized implementation for area, timing, and latency. Demonstrates expertise in hardware acceleration and low-level cryptographic implementations for secure data integrity.

TECHNICAL SKILLS

ML / AI: PyTorch, YOLOv8 (Ultralytics), EasyOCR, Transfer Learning, Fine-tuning, Data Augmentation, AdamW/RMSprop

Computer Vision: Object Detection, OCR Pipelines, Document Analysis, Image Processing, OpenCV

Languages: Python, PHP, Verilog, VHDL, Bash

Tools & Platforms: VS Code, Git/GitHub, Jupyter, Ansys HFSS, FPGA Tools, Ignition 8.3

Systems & Deployment: Linux, Docker, MySQL/SQL, OPC-UA, SCADA/HMI, Networking

EDUCATION

B.Tech in Electronics & Telecommunications (EXTC)

Shah & Anchor Kutchhi Engineering College | Expected May 2025 | SGPI: 6.66/10

HSC: ST. Mary's Convent High School (71.38%) | SSC: Jawaharlal Nehru Port Vidyalaya (65.00%)

CERTIFICATIONS & ACHIEVEMENTS

- AI Fluency: Framework & Foundations – Anthropic (2026)
- Claude with the Anthropic API – Anthropic (April 2026) | Cert. No: yfroa8pjaet9
- Ignition 8.3 Core Certification – Development & Troubleshooting
- AWS Academy Cloud Security Foundations
- Java Programming Certification | Introduction to Cybersecurity

LANGUAGES & INTERESTS

Languages: English (Professional) • Hindi (Native) • Marathi (Native)

Interests: Applied AI/ML, computer vision, model optimization, scalable deployment, and production ML systems.